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GEODESY

Module Name Vertical Control Using Digital Level

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Title	Geodesy			
Module Name	Vertical Control Using Digital Level			
Module Code				
Description	This module enables learners to describe vertical datum, levelling, various height systems and its related aspects. Develop and execute plan for height determination of unknown benchmark/point. Provide vertical control of required accuracy using Digital Level.			
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1.1 Basic Principle of Levelling

Levelling is one of the most fundamental operations in surveying.

It is used to determine the relative heights of points on the Earth's surface with respect to a vertical datum, usually Mean Sea Level (MSL).

Levelling forms the basis of:

- Vertical control surveys
- Engineering and construction works
- Drainage and irrigation planning
- Road, rail, and canal projects

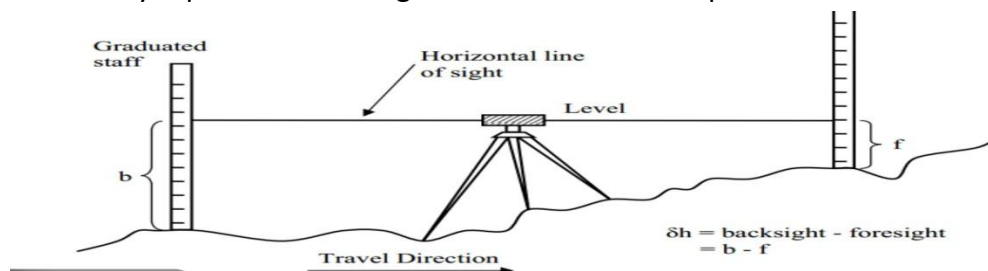
1.1.1 Definition of Levelling

Levelling is the process of measuring vertical distances between points by establishing a line of sight and observing vertical staff readings.

1.1.2 Basic Principle of Levelling

The basic principle of levelling is that the difference in elevation between two points is equal to the difference between their staff readings taken from a horizontal line of sight.

When the instrument's line of sight is horizontal, any difference in staff readings directly represents the height difference between points.



1.1.3 Components Involved in the Principle

Horizontal Line of Sight

The horizontal line of sight is the straight line seen through the telescope of the level instrument.

It is made horizontal by:

- a spirit bubble (in Auto / Spirit Levels), or
- an automatic compensator (in Digital Levels).

This line is adjusted so that it is level with the ground at that place, meaning it follows the natural level of the Earth caused by gravity.

Vertical Staff

- A graduated staff held vertically on survey points.
- Readings represent vertical distances from the horizontal line of sight.

1.1.4 Key Observations in Levelling

Backsight (BS)

- First staff reading taken on a point of known height (Benchmark).
- Used to establish the Height of Instrument (HI).

Foresight (FS)

- Staff reading taken on a point of unknown height.
- Used to compute its elevation.

1.1.5 Height Difference Computation

The difference in elevation between two points is calculated as:

Height Difference=Backsight (BS)–Foresight (FS)

- If $BS > FS \rightarrow$ Point is higher
- If $BS < FS \rightarrow$ Point is lower

1.1.6 Height of Instrument (HI) Concept

$HI = \text{Known RL (Reduced Level)} + BS$

$RL \text{ of New Point} = HI - FS$

This method ensures systematic and error-controlled height transfer.

1.1.7 Physical Meaning of Levelling

- Levelling follows the Earth's gravity field
- The horizontal line of sight corresponds to a level (equipotential) surface
- Hence, levelling provides physical heights, suitable for:
 - Water flow analysis
 - Engineering gradients
 - Drainage design

1.1.8 Importance of Levelling

Levelling ensures:

- Uniform height reference across projects
- Safe and efficient design of structures
- Correct slope and gradient control
- Reliable vertical control networks

1.2 Various components of levelling instruments

A Digital Level is an advanced levelling instrument that automatically reads, processes, records, and stores staff observations using electronic image-processing technology, eliminating manual reading errors associated with conventional optical levels.

Major Applications

- Precise and High-Precision Levelling (HPL)
- Vertical control surveys
- Benchmark establishment and transfer
- National and regional levelling networks
- Engineering deformation and settlement studies

Main Components of a Leica Digital Levelling Instrument

- Optical components
- Electronic components
- Mechanical components
- Control & interface components
- Data handling & storage components

- Power supply system
- Internal software & quality control system

All these components work together to produce accurate, repeatable, and reliable height measurements



- a) Optical sight
- b) Handle with integrated prism for viewing the circular level
- c) Circular level
- d) Touchscreen
- e) Battery compartment, also containing interface for USB stick and Mini USB
- f) Horizontal drive
- g) Push button to open battery compartment
- h) Function keys
- i) ON/OFF key
- j) Keyboard
- k) Eyepiece
- l) Protective cap for crosshair adjustment screw
- m) Horizontal circle

1.2.1 Optical Components

Objective Lens

The objective lens is the primary optical element of the digital level.

Functions:

- Collects light reflected from the barcode staff
- Forms a high-resolution image on the internal image sensor
- Ensures sharp and distortion-free imaging

Key characteristics:

- High optical clarity
- Minimal chromatic and spherical aberration
- High contrast, essential for barcode recognition

Unlike traditional optical levels, the observer does not visually read staff graduations.

Internal Focusing System

The internal focusing mechanism ensures that the barcode image is always sharp.

Features:

- Motorized or fine-focus adjustment
- Automatic or manual focusing
- Optimized for varying staff distances

Importance:

- Accurate decoding of barcode patterns
- Reliable distance and height computation

- Reduction of observation noise

1.2.2 Electronic Components

CCD / CMOS Image Sensor

The CCD/CMOS image sensor is the heart of a digital level.

Functions:

- Captures the optical image of the barcode staff
- Converts light signals into digital electrical signals
- Provides pixel-level accuracy

This component replaces the human eye used in conventional levelling.

Image Processing Unit

The image processing unit analyzes the captured staff image.

Functions:

- Decodes barcode patterns
- Automatically determines:
 - Staff reading (height)
 - Distance to staff
- Performs real-time internal quality checks

Advantages:

- Eliminates reading and booking errors
- Provides consistent results independent of observer skill

Electronic Compensator

The electronic compensator ensures that the line of sight remains horizontal.

Functions:

- Automatically corrects small tilts of the instrument
- Maintains a true horizontal line of sight
- Operates within a defined tilt range

Importance:

- Directly controls accuracy
- Ensures stability during observation
- Critical for first-order and high-precision levelling

Compensator performance defines the reliability of a digital level.

1.2.3 Mechanical Components

Telescope Assembly

The telescope assembly integrates:

- Objective lens
- Image sensor
- Internal optics

Functions:

- Provides magnified view for aiming
- Contains reticle/crosshair for approximate staff alignment

- Houses optical-electronic integration

Levelling Base (Tribrach / Foot Screws)

The levelling base supports and stabilizes the instrument.

Components:

- Three precision foot screws
- Rigid base plate

Functions:

- Used for coarse levelling
- Ensures compensator operates within its working range

Circular Bubble

The circular bubble is used for preliminary levelling.

Functions:

- Provides quick coarse levelling
- Ensures the instrument is approximately horizontal before measurement

1.2.4 Barcode Staff (Essential External Component)

Barcode Levelling Staff

A barcode staff is mandatory for digital levelling.

Features:

- Black-and-white coded bar pattern
- Designed for automatic image recognition
- High contrast and precision printing

Advantages:

- Automatic reading
- Elimination of parallax and human reading errors
- Faster observations



Invar Staff

For high-precision work, Invar staffs are used.

Characteristics:

- Made of Invar alloy (very low thermal expansion)
- Maintains length stability under temperature variations

Applications:

- First-order levelling
- National height networks
- Scientific and deformation studies



Staff calibration is mandatory for precision levelling.

1.2.5 Control & Interface Components

Display Screen

The digital level is equipped with a high-contrast LCD.

Displays:

- Staff reading

- Distance to staff
- Observation quality indicators
- Job, line, and point information

Keypad / Control Buttons

The keypad allows user interaction.

Functions:

- Job creation and management
- Observation triggering
- Menu navigation
- Data review

1.2.6 Data Handling & Storage Components

Internal Memory

Internal memory stores all survey data.

Stored information:

- Staff readings
- Distances
- Point IDs
- Heights and observations

Advantages:

- Eliminates manual field books
- Reduces transcription errors

USB / Data Transfer Port

Used for data export and transfer.

Supported formats:

- ASCII

Purpose:

- Data processing
- Archival
- Network adjustment

1.2.7 Power Supply System

Rechargeable Battery

Features:

- Long operating life
- Suitable for full-day field work
- Stable voltage supply

Battery Compartment

Characteristics:

- Secure and weather-sealed
- Quick battery replacement in the field

1.2.8 Environmental & Protection Features

- Dust-resistant housing
- Water-resistant design
- Suitable for harsh climatic conditions
- Reliable in long-duration surveys

1.2.9 Internal Software Components

Levelling Programs

Built-in programs include:

Line levelling

BFFB / FBBF observation sequences

Check levelling

High-precision levelling modes

Error & Quality Control Algorithms

The software automatically:

- Rejects poor observations
- Checks:
 - Staff tilt
 - Focus quality
 - Compensator operating range
- Ensures consistent accuracy

1.3 Testing and adjustment of levelling instrument.

Testing and adjustment of a digital levelling instrument are essential to ensure that the instrument provides accurate, reliable, and repeatable height measurements. Although digital levels reduce human errors, they are still affected by instrumental, mechanical, and electronic imperfections.

Purpose of Testing and Adjustment

The main objectives are:

- To ensure the line of sight is truly horizontal
- To detect and minimize collimation error
- To verify the correct functioning of the electronic compensator
- To confirm reliable barcode staff reading
- To maintain accuracy standards prescribed for:
 - Secondary levelling
 - Precise levelling
 - High-precision levelling

Testing identifies the presence of error

Adjustment removes or compensates for the detected error

Errors Relevant to Digital Levelling Instruments

Digital levelling instruments are mainly affected by:

- Collimation error

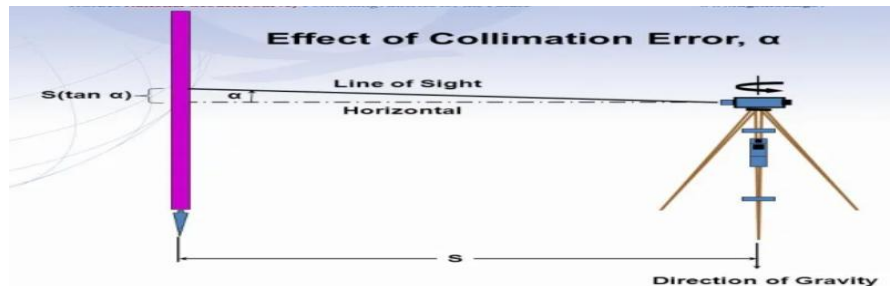
- Compensator malfunction
- Staff reading and calibration error
- Electronic or software instability

Unlike conventional levels, **bubble tube errors are minimal**, as digital levels use electronic compensation.

1.3.1 Collimation Error in Digital Levels

Collimation error occurs when the optical–electronic line of sight is not horizontal, even though the compensator indicates that the instrument is level.

This is the most critical error in both optical and digital levelling.



Two-Peg Test (Primary Test for Digital Levels)

Principle

- When the instrument is placed midway between two points, the collimation error cancels out.
- When the instrument is placed near one point, the error becomes evident.

Field Procedure (Digital Level)

Select two firm points **A** and **B**, about **60 m apart**.

Place the digital level **midway** between A and B.

Take barcode staff readings: a_1 at A and b_1 at B

Compute the true height difference: $\Delta h = a_1 - b_1$

Shift the instrument **close to point A**.

Take readings again: a_2 at A and b_2 at B

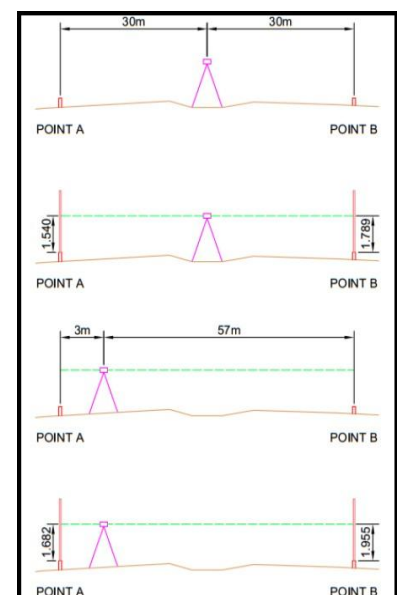
Compute: $\Delta h' = a_2 - b_2$

Interpretation

- If $\Delta h = \Delta h' \rightarrow$ No collimation error
- If $\Delta h \neq \Delta h' \rightarrow$ Collimation error present

Adjustment in Digital Levels

- User cannot mechanically adjust collimation in digital levels.
- Correction is done by:
 - Internal software calibration, or
 - Authorized service center adjustment



1.3.2 Compensator Testing

Role of Compensator

The compensator automatically maintains a horizontal line of sight by correcting small tilts of the instrument.

Compensator Test Procedure

- Set up the digital level normally and level it.
- Take a staff reading.
- Gently tap the instrument or tripod.
- Observe the reading on the display.

Interpretation

- Reading returns to the same value : Compensator working correctly
- Reading fluctuates or drifts : Compensator faulty

A faulty compensator severely degrades levelling accuracy.

1.3.3 Barcode Staff Testing

Staff Verticality Check

- Ensure the staff is held truly vertical using:
- Circular bubble on staff

Tilted staff leads to systematic height error.

Staff Calibration Check

Invar barcode staffs must be:

- Periodically calibrated
- Free from bends and mechanical damage

1.3.4 Distance Measurement Consistency Test

Digital levels also compute distance from barcode patterns.

Procedure:

- Observe the same staff position repeatedly
- Distance values should remain consistent

Variation indicates:

- Poor focus
- Staff damage
- Electronic instability

1.3.5 Software & Display Check

Before field work:

- Check display clarity
- Verify correct levelling mode (Line / HPL)
- Ensure correct observation sequence (BFFB / FBBF)
- Confirm memory availability

1.3.6 Routine Field Checks (Daily Practice)

- Instrument stability on tripod
- Circular bubble centered

- Compensator free movement
- Staff cleanliness and readability
- Battery charge level

Testing and adjustment of a digital levelling instrument ensure correct horizontality, reliable compensator action, and accurate barcode staff readings, thereby maintaining the required accuracy for precision and high-precision levelling.

1.4 The sources of error in levelling and its adjustments.

Levelling is a high-accuracy surveying operation used to determine differences in elevation between points.

Even with modern instruments (auto levels and digital levels), levelling observations are affected by various sources of error.

If these errors are not identified and adjusted, they may lead to:

- Incorrect benchmark elevations
- Poor vertical control
- Failure of engineering designs (drainage, canals, roads)

Hence, understanding sources of error and their adjustments is essential.

Classification of Errors in Levelling

- Instrumental Errors
- Personal (Observational) Errors
- Natural (Environmental) Errors

1.4.1 Instrumental Errors and Their Adjustments

Instrumental errors arise due to imperfections or maladjustments of the levelling instrument.

Collimation Error

- Line of sight is not truly horizontal when the instrument is levelled.
- Produces systematic error that increases with sight length.

Adjustment

- Perform Two-Peg Test regularly
- Adjust collimation (optical levels)
- For digital levels: software calibration / service adjustment
- Keep backsight and foresight distances equal

Bubble Tube / Compensator Error

- Bubble axis not perpendicular to vertical axis (optical level)
- Faulty compensator (auto/digital level)
- Line of sight not horizontal

Adjustment / Remedy

- Test and adjust bubble tube
- Perform compensator test
- Service instrument if compensator is unstable

Staff Graduation Error

- Incorrect or worn staff graduations
- Uncalibrated barcode staff
- Systematic height error

Adjustment / Remedy

- Use calibrated staff
- Apply staff correction
- Periodic laboratory calibration (mandatory for precision levelling)

1.4.2 Personal (Observational) Errors and Their Adjustments

Personal errors occur due to **human limitations or carelessness**.

Staff Not Held Vertical

- Staff tilted forward or backward
- Observed reading becomes **greater than true reading**

Adjustment / Remedy

- Use circular bubble on staff
- Rock the staff gently and take minimum reading

Incorrect Focusing (Parallax Error)

- Improper focusing of eyepiece or objective
- Inaccurate staff readings

Adjustment / Remedy

- First focus eyepiece on crosshairs
- Then focus objective on staff image

Booking and Reading Errors

- Wrong reading entry
- Transposition of BS and FS
- Manual booking mistakes
- Gross levelling errors

Adjustment / Remedy

- Proper field-book format
- Immediate arithmetic checks
- Use digital levels to eliminate booking errors

1.4.3 Natural (Environmental) Errors and Their Adjustments

Natural errors arise due to atmospheric and Earth-related factors.

Curvature of the Earth

- Earth is curved, not flat
- Line of sight is horizontal but Earth's surface curves away
- Staff appears lower at long distances

Correction: $C_c = -0.0785d^2$ (meters)

Adjustment / Remedy

- Keep sight lengths short

Balance backsight and foresight distances

Atmospheric Refraction

- Bending of light due to temperature and air density variation
- Line of sight bends downward
- Partially compensates curvature

Correction: $C_r = +0.0112d^2$

Combined Curvature & Refraction

$C = -0.0673d^2$ $C = -0.0673$

(Where d is distance in km)

Practical Remedy

- Equal distance of BS and FS (errors cancel automatically)

Wind, Heat & Vibration

- Staff movement due to wind
- Heat shimmer
- Instrument vibration

Adjustment / Remedy

- Avoid levelling during hot afternoons
- Shield staff from wind
- Use firm tripod setup

1.4.4 Adjustments in Levelling (Error Control Methods)

Field Adjustments

- Equalize BS and FS distances
- Short sight lengths (30–60 m)
- Frequent instrument testing
- Stable turning points

Computational Adjustments

- Arithmetic check: $\sum BS - \sum FS = \text{Last RL} - \text{First RL}$
- Distribute closing error proportionally to distances

Errors in levelling arise due to instrumental imperfections, observer mistakes, and natural causes, and they are minimized by proper testing, balanced sight lengths, staff calibration, and suitable field procedures.

1.5 Understanding the Vertical Datum Concept

A vertical datum represents a precisely defined reference surface from which all heights and depths are systematically measured. Think of it as the fundamental "starting line" for elevation. Without this agreed upon zero point, comparing heights between different locations would be impossible.

This reference serves as the basis of consistent and accurate elevation data across diverse fields including surveying, mapping, cartography, and engineering applications.

Every time we measure how high a mountain peak stands or how deep an ocean trench extends, we are referencing to a vertical datum.

1.5.1 Vertical Datum as a Reference Surface

Zero Level reference: Functions as the universally agreed "zero level" from which all vertical measurements begin, whether measuring upwards into mountains or downwards into ocean depths.

Surface Types: Can manifest as either a physical surface-such as mean sea level observed at tide gauges or a sophisticated mathematical model like the geoid that accounts for gravitational variations.

Global Consistency: Enables meaningful comparison of heights anywhere on Earth, overcoming the challenge posed by our planet's irregular, non-spherical shape and varying gravitational field.

1.5.2 Types of Vertical Datums

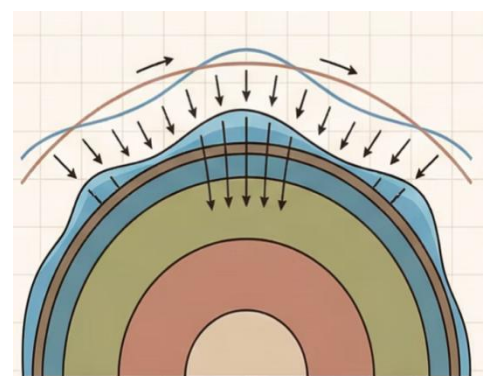
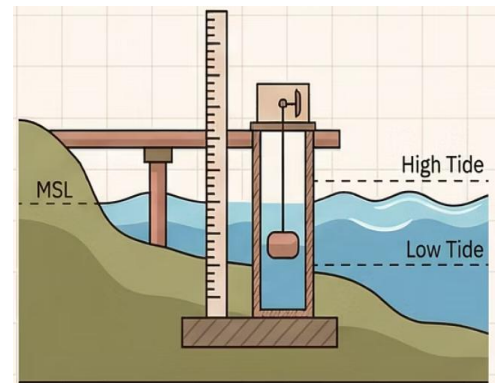
Different applications require different reference systems. Vertical datums fall into three primary categories, each optimised for specific measurement contexts and technological approaches.

Tidal Datums: Founded upon careful sea level observations recorded over extended periods at coastal tide gauge stations. Mean Sea Level (MSL) represents the most familiar example, calculated by averaging tidal heights over the period of about 19 years to eliminate fluctuations caused by various factors.

- Based on actual water surface measurements
- Ideal for coastal engineering and navigation
- Location-specific to tide gauge stations

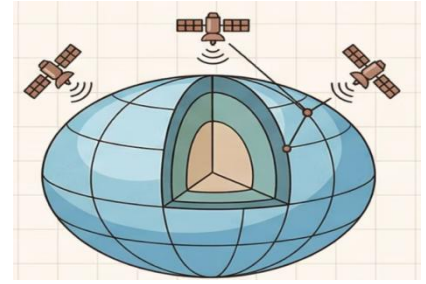
Geoid: Geoid is an equipotential surface closely follows the sea level representing Earth's shape as defined purely by gravity. This approach provides a physically meaningful reference that accounts for gravitational variations across the planet. Based on actual water surface measurements Ideal for coastal engineering and navigation Location-specific to tide gauge stations

- Represents gravitational equipotential surface
- Provides consistent global reference



- Essential for precise scientific applications

Geometric Datums: Built upon mathematical ellipsoid models, smooth, geometric approximations of Earth's shape. These are fundamental to satellite positioning systems like GNSS, which measure distances from satellites to Earth's surface.

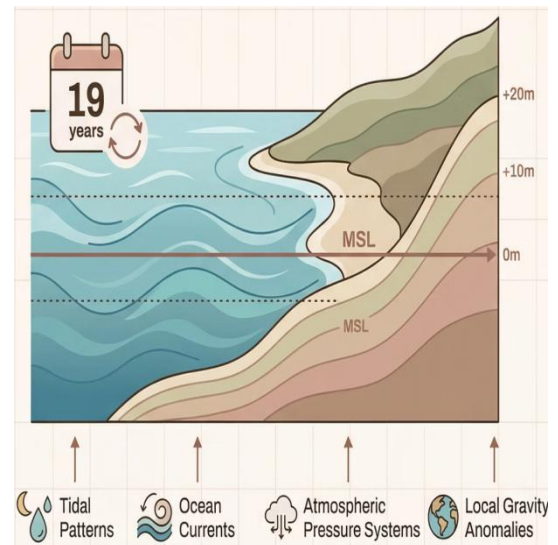


- Simple mathematical definition
- Optimised for satellite positioning
- Requires transformation to other datum types

1.5.3 Mean Sea Level as a Vertical Datum

Mean Sea Level (MSL) stands as perhaps the most intuitively understood vertical datum. It represents the average height of the ocean's surface, meticulously calculated over a standardised 19-year tidal cycle for a period chosen to capture and average out all major tidal variations and astronomical influences.

This long observation period ensures that short-term fluctuations caused by storms, seasonal changes, and lunar cycles are smoothed out, revealing the true mean position of the sea surface. MSL serves as the common zero elevation reference for measuring land heights in many national surveying systems worldwide.



However, MSL is not uniform globally. Regional variations arise from complex interactions of tides, ocean currents, atmospheric pressure systems, water temperature and density differences, and local gravity anomalies. The sea surface off Japan, for instance, sits at a different absolute height than off Brazil, even when both are measured relative to the same reference ellipsoid.

1.5.4 The Geoid: Earth's Gravity-Based Reference

Geoid models are constructed using satellite gravity measurements, terrestrial gravity observations, and sophisticated mathematical techniques. Modern models like EGM2008 represent decades of data collection and can resolve geoid variations down to remarkably fine spatial scales

Physical Definition The geoid represents the "true" shape that Earth's mean sea level would assume under the influence of gravity alone, extended conceptually through the continents. It's an equipotential surface⁴ meaning gravitational potential is identical everywhere on this surface

Irregular Surface Unlike the smooth, mathematically perfect ellipsoid, the geoid undulates irregularly due to variations in Earth's mass distribution. Mountains,

ocean trenches, and density variations in the crust all influence local gravity, creating "bumps" and "dips" in the geoid

Meaningful Reference The geoid provides a physically meaningful zero elevation surface that aligns with how we intuitively understand "level"4water naturally flows perpendicular to the geoid surface. This makes it the ideal reference for vertical datum in scientific and engineering contexts

1.5.5 Ellipsoid-Based Vertical Datums

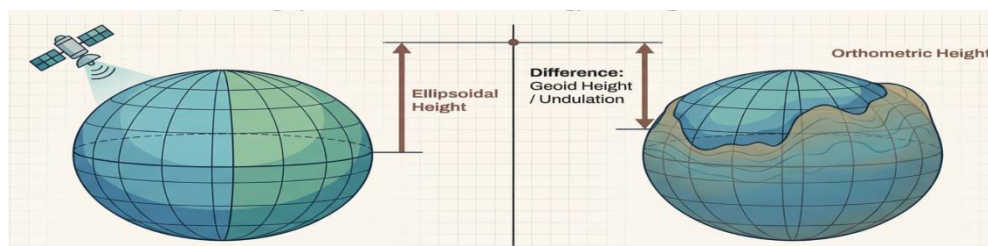
Mathematical Simplicity ellipsoids are mathematically smooth surfaces that approximate Earth's overall shape essentially a slightly flattened sphere defined by precise geometric parameters. Unlike the irregular geoid, an ellipsoid can be described by just a few numbers: equatorial radius, polar radius, and flattening factor

This mathematical simplicity makes ellipsoids ideal for satellite positioning calculations. Global Navigation Satellite Systems (GNSS) like GPS naturally measure distances and compute positions relative to a reference ellipsoid, as the satellite orbit calculations and signal processing are much simpler using geometric models

GNSS Positioning Satellites compute your position relative to an ellipsoid reference surface

Height Transformation Ellipsoidal heights must be converted using geoid models to obtain meaningful elevations

Practical Application Final heights referenced to geoid or MSL for engineering and mapping use (a height measured from an ellipsoid differs from a height measured from the geoid. The separation between these surfaces called the geoid height or geoid undulation it can exceed 100 metres in some locations, making transformation between systems essential for practical applications.)



1.5.6 Practical Importance of Vertical Datum

The abstract concept of a vertical datum translates into tangible real-world impact across numerous critical applications. Consistent elevation data isn't merely convenient, it's essential for safety, functionality, and coordination of human activities.



Construction & Engineering Ensures building foundations are level, roads maintain proper grades for drainage, and structures connect seamlessly across project boundaries

Flood Risk Assessment Determines which areas face inundation risk by comparing terrain elevation with predicted water levels during storm events or sea level rise

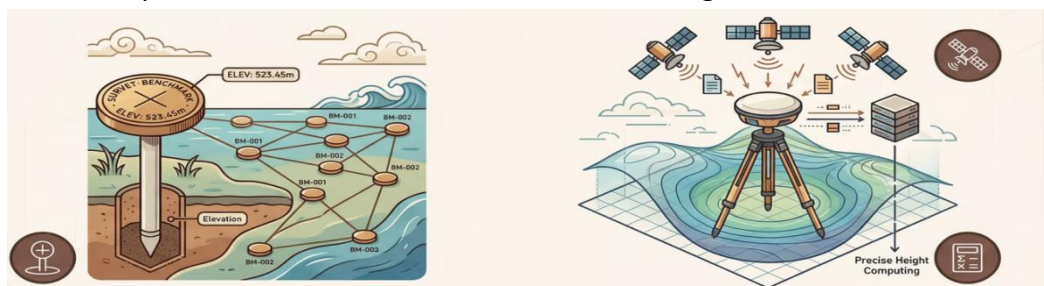
Maritime Navigation Establishes water depths for safe passage and bridge clearances for vessels, preventing groundings and collisions with overhead structures

Infrastructure safety Vertical datums prove critical for infrastructure safety calculations. Determining flood risk for a hospital, calculating whether a bridge provides adequate clearance for river traffic, or assessing whether an airport runway has proper drainage all depend on accurate, consistent elevation data referenced to an appropriate vertical datum.

Data Integration Perhaps most importantly, vertical datums allow integration of elevation data from disparate sources historical surveys, modern GNSS measurements, LIDAR scans, and satellite observations can all be brought into a common reference frame, enabling comprehensive spatial analysis and decision-making.

1.5.7 Vertical Datum Realisation

Establishing a vertical datum in the physical world4making it practically usable rather than merely theoretical4requires a network of precisely measured reference points and modern measurement technologies.



Physical Benchmarks Permanent survey markers with precisely determined elevations are installed throughout a region4typically brass or bronze discs set in bedrock or stable structures, each with a unique identifier and published elevation value.

Levelling Networks Traditional spirit levelling techniques establish relative height differences between benchmarks with millimetre precision, building networks that span entire continents through decades of careful surveying work.

GNSS Integration Modern approaches combine Global Navigation Satellite System observations with detailed geoid models, allowing rapid determination of heights without requiring physical connections to existing benchmarks

Continuous Monitoring Networks of continuously operating GNSS stations track subtle land movements, subsidence, and tectonic shifts, enabling detection of changes that might affect vertical datum stability over time.

1.6 Historical development of vertical datum i.e. MSL (Mean sea Level), Geoid

India's journey toward a unified and scientific vertical datum spans nearly two centuries. It begins with tide gauges and spirit levelling, grows through massive geodetic campaigns, and culminates today in the recognition that a modern nation cannot depend on local Mean Sea Levels alone.

This chapter explains how India's height systems evolved, why different datums existed across time, and why a shift to a geopotential-based national datum (IVD) is now essential.

1.6.1 Before 1858 – India had NO scientific vertical datum

This is exactly what the early part reveals:

- Heights were determined only from reciprocal vertical angles between trig stations.
- But refraction was unpredictable → errors of several feet.
- On the plains of North India, even a 1–2 ft error mattered for canals and irrigation.
- Sometimes reciprocal visibility was not even simultaneous — proving the method unreliable.

1.6.2 1858 – Birth of Precise Levelling in India (Foundations of a Datum)

- 1858 marks the start of spirit levelling in India.
 - First line: Karachi → Attock → Dehra Dun → Sirmur (closely following the GTS triangulation).
 - This was the first time instrument & staff replaced vertical angles.
- “India still did not have a vertical datum. We only had heights relative to each other, not relative to the Earth.”

1.6.3 1875–1880 – General J.T. Walker's Vision:

This is the turning point in India's vertical datum history — beautifully documented in Volume XIX. General Walker realised:

- A vertical datum must begin at the sea, not inland.
- Different parts of India were using local height systems → confusion.
- He imagined a continuous chain of levelling lines tied to tidal observatories.

A Chain of Tidal Observatories Around India's Coast

- Karachi
- Bombay
- Cochin
- Madras
- False Point
- Beypore
- Vishakhapatnam
- Nagapattinam

Each observatory supplied a Mean Sea Level (MSL) determination, which became the starting reference for levelling.

This marks India's first scientifically established vertical reference system.

"A vertical datum was born when levelling touched the sea and carried that reference inland."

1.6.4 1858–1909 – Building the Indian Vertical Datum (The Great Levelling Era)

Over 50 years, the Survey of India constructed:

- 86 main precise levelling lines
- Thousands of branch lines
- Tens of thousands of benchmarks on masonry, rock, railway stations, bridges, culverts
- This created India's first continent-wide vertical control network, all tied to MSL from tidal observatories.
- This produced the 1909 Indian Vertical Datum (IVD1909) — India's first national reference for heights.

1.6.5 The 1909 datum assumed

"Mean Sea Level at all nine TG stations represents the same level surface." But modern oceanography shows this was incorrect.

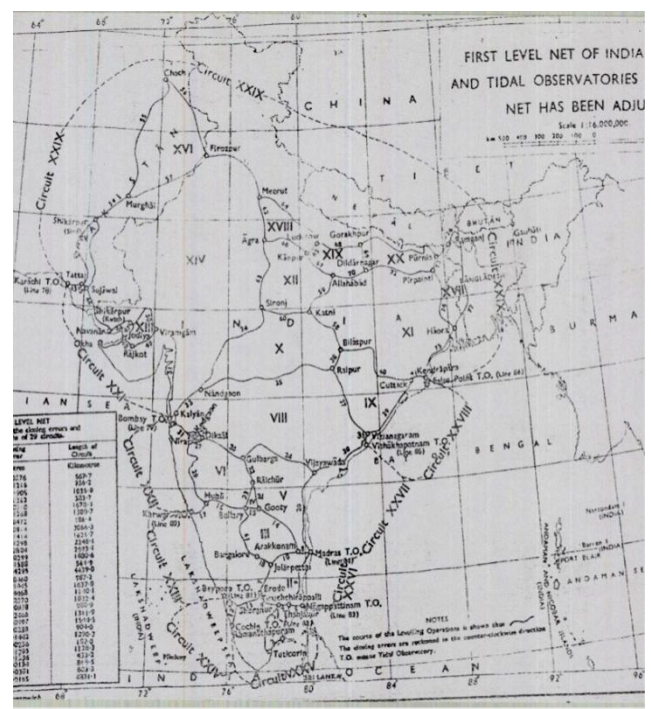
MSL varies due to:

- Dynamics of ocean
- Temperature & salinity
- Currents
- Winds
- Local sea surface topography (SST)

Thus, the original IVD1909 was not a true equipotential surface.

The Hidden Weaknesses of the Old Indian Vertical Datum

"MSL is not an equipotential surface few metres." This means the "zero" itself was fundamentally flawed.



Sea Level Variations : The nine tide gauges used in 1909 did not represent a uniform sea surface.

- Levelling distortions accumulated for 100 years
- Old lines had systematic errors.
- Earth's crust moves over time.
- Benchmarks sink or shift.

1.6.6 Modernization Begins – 2006 to 2009

Triggered by the issues above, Survey of India initiated a scientific overhaul.

From RIVD 2009:

New High-Precision Levelling Framework (2006–2008)

- 20,000 km of new levelling
- Benchmarks every 25–30 km
- Gravity measured at each BM(to compute geopotential numbers)

GNSS observations at 8 tide gauges This allowed:

- Ellipsoidal heights
- New geoid calculations
- Evaluation of sea surface topography at each Tide Gauge

Determination of a New Geopotential Value (W_0)

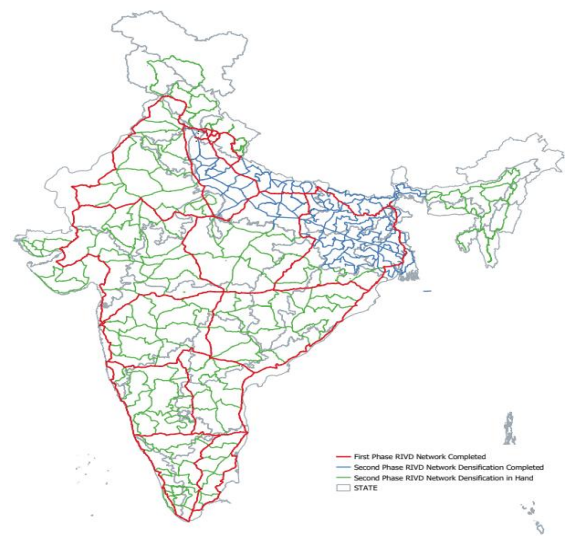
- Local geoid potential
- Consistency with global W_0
- Equipotential surface definition

This turned height definition from MSL-based to physics-based a major scientific leap.

Redefinition of Indian Vertical Datum (IVD2009). This is India's modern height reference system.

1.6.7 Key characteristics (from RIVD 2009):

- Geoid / equipotential surface replaces MSL(MSL used only as an observation input, not as the datum itself)
- Datum defined in terms of geopotential $W_0 \rightarrow$ making it compatible with GNSS
- Helmert Orthometric Heights adopted nationwide
- Integration with ITRF-based Indian Geodetic Datum
- Future-ready for gravimetric geoid-based heighting



1.7 Classification of levelling (ST/DT, HP) and their Accuracy standards

Depending upon the type of instrument used, procedure adopted for observations and refinement of computations, the leveling is classified into the following categories:-

- **High Precision**
- **Precision Leveling**
- **Secondary Leveling**
- **Tertiary Leveling**

1.7.1 High Precision Levelling

is the most accurate form of spirit levelling, carried out to establish or refine the national vertical datum and for scientific geodesy.

Main Features

- Double run (forward + backward)
- Digital levels (DNA-03, Dini)
- Invar staffs only
- Equal sight lengths (very strict)
- Actual observed gravity values used
- Computation in geopotential numbers

Purpose

- Redefinition of vertical datum (e.g. RIVD)
- Geoid & gravity studies
- Crustal deformation, sea level studies

Accuracy : $4 \sqrt{K}$ mm

1.7.2 Precision Levelling :

Precision levelling is carried out to extend reliable height control from primary datum points to regional networks.

Main Features

- Double run (forward + backward)
- High-quality optical or digital levels
- Invar staff preferred
- Normal gravity generally used
- Strict field procedures (but less strict than HPL)

Purpose

- Regional height control
- Important engineering projects
- Control for mapping & infrastructure

Accuracy : $4\sqrt{K}$ mm

1.7.3 Secondary Levelling :

Secondary levelling is used to further densify the levelling network for administrative and engineering needs.

Main Features

- Usually double run
- Ordinary precise levels
- Wooden or aluminium staff

1.7.4 The Tertiary leveling

Depending on the purpose of survey, extent of work, and required accuracy, Tertiary levelling operations are classified into Two types.

Double Tertiary Levelling: in which the permissible error is $12\sqrt{K}$ mm where K is the distance in km. Agreement between two set of observation at a station is 0.002m

Single Tertiary Levelling: in which the permissible error is $24\sqrt{K}$ mm where K is the distance in km

1.8 Methods of Levelling

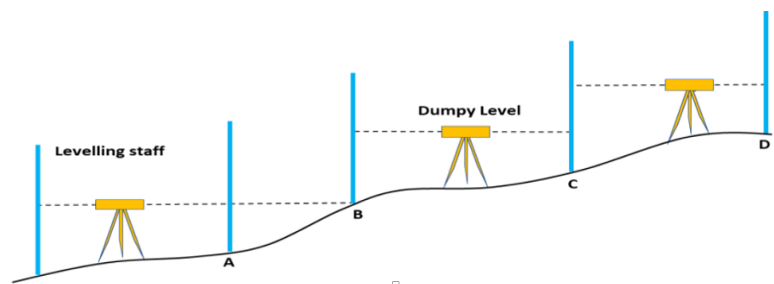
1.8.1 Differential Levelling

Determines the difference in elevation between two or more points that are far apart, not visible from each other.

It is the most commonly used method in construction and topographic works.

Principle:

Elevation of an
unknown point =
Elevation of
known point +
Back Sight (BS) –
Fore Sight (FS)



Field Procedure:

- Set up the instrument midway between two points.
- Take BS on the known point (BM).
- Take FS on the next point.
- Move instrument and repeat until the destination point is reached.
- Apply arithmetic check:
 $\Sigma BS - \Sigma FS = \text{Last RL} - \text{First RL}$

Applications:

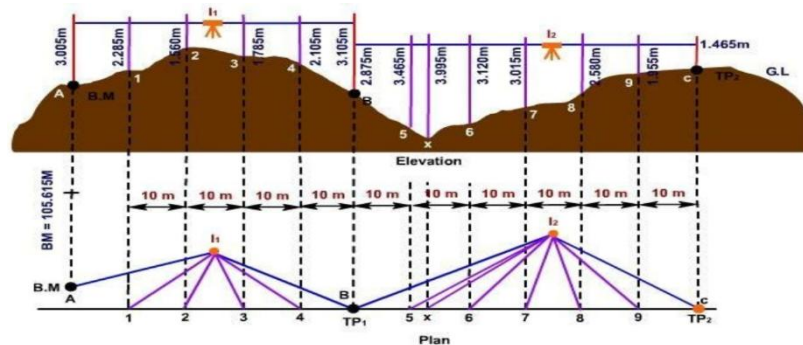
- Road, canal, and pipeline surveys.
- Level transfer in hilly terrains.
- Construction of long corridors or linear projects.

1.8.2 Profile Levelling (Longitudinal Levelling)

Used to determine the elevation of points along a selected line (alignment) — typically the centre line of a road, railway, canal, or pipeline.

Objective:

To draw the longitudinal section (profile) of the ground surface to design gradients, slopes, and cut/fill sections.



Procedure:

- Set out alignment using pegs at fixed intervals (e.g., every 10 m or 20 m).
- Take BS on the starting BM and IS (Intermediate Sights) on all points along the line.
- Plot RLs and create a profile chart showing natural ground level (NGL) and proposed formation level (FL).

Applications:

- Road, railway, and canal alignments.
- Drainage or sewer design.
- Pipeline and tunnel planning.

1.8.3 Cross-Section Levelling

Determines the ground elevations on both sides of the main survey line (longitudinal line).

These are taken perpendicular to the alignment at regular intervals.

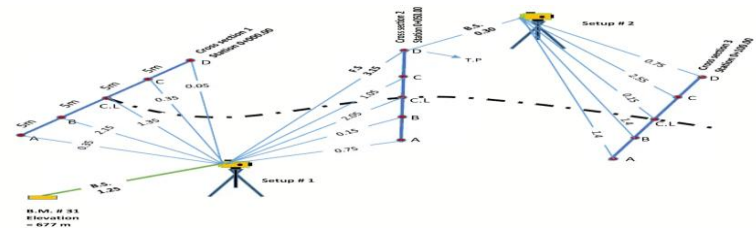
Purpose: To calculate earthwork quantities and understand the terrain's cross-sectional shape.

Procedure:

- After completing profile levelling, take staff readings at specified offsets (e.g., 5 m, 10 m, 15 m) left and right of the centre line.
- Plot cross-sections on graph paper to scale.
- Compute areas using trapezoidal or Simpson's rule for volume estimation.

Applications:

- Road and canal earthwork computation.
- Volume estimation for embankment or cutting.
- Contour and terrain representation.



1.8.4 Reciprocal Levelling

Used when two points are inaccessible (e.g., across rivers, valleys, or obstacles) where normal levelling cannot be performed from a single setup.

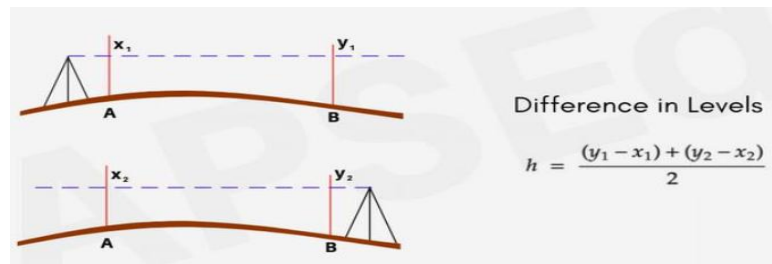
Principle:

By observing reciprocal readings from both banks, errors due to curvature, refraction, and collimation are eliminated.

Procedure:

- Set instrument near point A; take BS on A and FS on B.
- Move instrument near B; take BS on B and FS on A.
- Calculate true difference of level using the average of corrected observations.

Formula:
True



difference of level = $[(y_1 - x_1) + (y_2 - x_2)] / 2$
(Where x_1, y_1, x_2, y_2 are readings from both setups)

Applications:

- Levelling across rivers, valleys, or large ponds.
- Transfer of elevation to inaccessible locations.
- Checking collimation error of instrument.

1.9 Accuracy Standards

Accuracy in levelling refers to the degree of closeness between the *measured elevations* and their *true values*.

Because levelling involves cumulative readings over long distances, even small errors — such as 1 mm per setup — can produce significant total misclosures if not controlled properly.

To ensure uniform precision and reliability, professional standards define tolerable limits of misclosure (difference between observed and true elevation). These limits depend on the type of survey, distance covered, and instrument precision.

The most widely used rule for controlling misclosure is the $\sqrt{K}$ (square root of distance) rule.

1.9.1 The $\sqrt{K}$ Rule

The permissible or tolerable error in levelling is proportional to the square root of the total distance levelled.

Formula: Allowable Misclosure (mm) = $C \times \sqrt{K}$

Where:

- C = Constant depending on type of levelling (precision level)
- K = Total distance levelled (in km)
- Misclosure = Difference between observed and theoretical elevation closure

This formula recognizes that random errors increase with the square root of the distance (not linearly), hence ensuring practical tolerance limits.

Standard Accuracy Classes

Type of Survey	Tolerable Misclosure ($\sqrt{K}$ Rule)	Remarks
Primary Levelling	$\pm 4 \text{ mm } \sqrt{K}$	Used for geodetic control, national benchmarks, and scientific monitoring
Secondary Levelling	$\pm 8 \text{ mm } \sqrt{K}$	Used for project-level and engineering works such as roads, canals, pipelines
Double Tertiary Levelling	$\pm 12 \text{ mm } \sqrt{K}$	Used for temporary control or quick benchmark transfers
Single Tertiary Levelling	$\pm 24 \text{ mm } \sqrt{K}$	Used for Local work

Primary Levelling

High-accuracy levelling operations carried out for national, regional, or scientific control networks, requiring errors within a few millimetres over long distances.

Key Features:

- Observations taken in both directions (double-run levelling).
- Balanced sight lengths and equal setups.
- Use of digital or automatic precise levels with invar staffs.
- Observation under calm atmospheric conditions (early morning or late evening).
- Temperature and curvature/refraction corrections applied.

Tolerable Misclosure: $\pm 4 \text{ mm } \sqrt{K}$

(e.g., for 16 km of levelling: $\pm 4 \times \sqrt{16} = \pm 16 \text{ mm}$)

Applications:

- National geodetic networks (Survey of India, GTS).
- Deformation studies of dams, bridges, and large structures.
- Tidal and hydrological observations.

Secondary Levelling

General-purpose levelling used in engineering and construction projects for establishing design elevations, gradients, and volume estimation.

Key Features:

- Performed using Auto Level or Digital Level.
- Should be used in subdividing loops of 1st-order leveling
- Intermediate sights (IS) taken as required for ground profile.
- Second-order lines should not be extended more than 20 km from lines of the first-order; they may be single-run lines but must always be loops or circuits closed upon lines of First order.
- Acceptable for large-scale mapping and civil works.

Tolerable Misclosure: $\pm 8 \text{ mm} \sqrt{K}$

(e.g., for 9 km of levelling: $\pm 8 \times \sqrt{9} = \pm 24 \text{ mm}$)

Applications:

- Highway and railway alignment surveys.
- Irrigation and drainage design.
- Town planning and site development.
- Pipeline, sewer, and water supply projects.

Tertiary Levelling

Rapid levelling conducted to transfer elevations from one known benchmark to another, typically as a preliminary or auxiliary operation.

Key Features:

- Third-order lines should not be extended more than 50 km from lines of the first- or second-order; they may be single-run lines but must always be loops or circuits closed upon lines of equal or higher order.
- Objective is quick transfer of height, not high accuracy.
- Instrument set-up time minimized; distances between stations can be longer.

Tolerable Misclosure: $\pm 12 \text{ mm} \sqrt{K}$ for DT

$\pm 24 \text{ mm} \sqrt{K}$ for ST

Applications:

- Establishing temporary benchmarks (TBM) for short-term construction.
- Preliminary route surveys and reconnaissance mapping.
- Checking approximate gradient for feasibility.

Misclosure Computation Example

Given: Distance = 9 km; Survey Type = Tertiary Levelling

Measured Misclosure = 0.045 m

Allowable Misclosure = $\pm 12 \text{ mm} \times \sqrt{9} = \pm 36 \text{ mm} = \pm 0.036 \text{ m}$

Since measured misclosure (0.045 m) > allowable (0.036 m),

Survey does not meet accuracy standard.

Must repeat observations or adjust systematically after identifying cause.

1.9.2 Adjustment of Misclosure

When the total misclosure is within tolerance, it may be distributed proportionally along the levelling line to correct each station's RL.

Formula: Correction at any point = $\frac{\text{Error} \times \text{Distance to that point}}{\text{Total Distance}}$

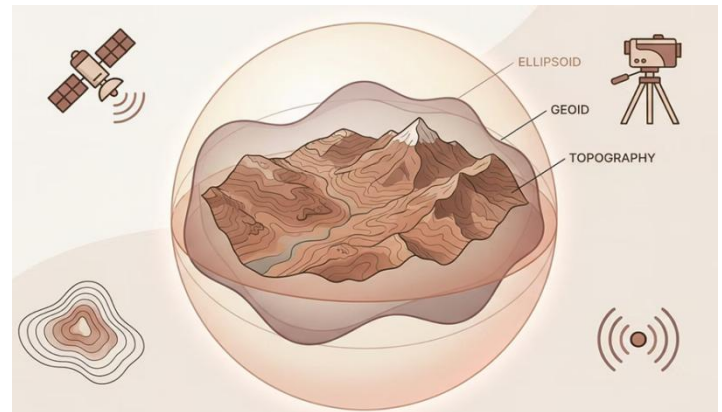
This ensures uniform distribution of residual error along the line, maintaining elevation consistency.

1.10 Various Height systems i.e., Orthometric, Ellipsoidal and Geopotential number etc

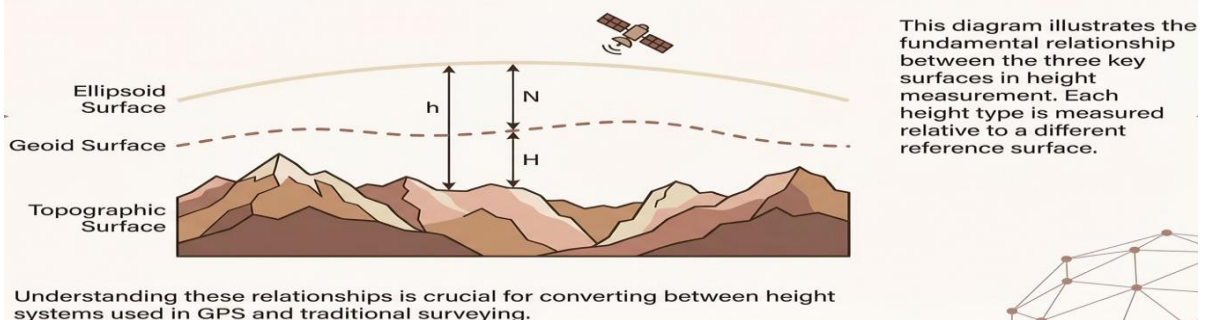
Height systems provide a one-dimensional coordinate to express vertical position relative to a reference surface. Different height systems exist because the Earth's shape and gravity field are complex and vary spatially across the globe.

Accurate height measurement is crucial for surveying, mapping, navigation, and engineering projects worldwide. From constructing bridges to predicting flood risk, understanding vertical position is fundamental to modern infrastructure and scientific research.

Surveying: Precise elevation data for construction and land development projects



Visualising the Three Surfaces



Mapping: Creating accurate topographic maps and digital elevation models

Navigation: Aviation altitude references and maritime depth measurements

Engineering: Infrastructure design requiring exact vertical positioning

1.10.1 Ellipsoidal Height: The Geometric Reference

Ellipsoidal height is defined as the vertical distance from a point on the Earth's surface to a mathematically defined reference ellipsoid. The ellipsoid is a smooth, regular shape approximating the Earth's figure, such as the widely used WGS 84 ellipsoid.

Direct GNSS Measurement: Ellipsoidal height is what GPS and GNSS receivers directly measure without requiring additional corrections or gravity data.

Geometric Nature: It is a purely geometric height, not accounting for gravity variations or actual sea level across different locations.



Practical Example: A GPS device reports an elevation of 150 m above the WGS 84 ellipsoid at a given location, representing the geometric distance upward.

1.10.2 Geopotential Number: Linking Height to Gravity Potential

The geopotential number quantifies the difference in gravitational potential energy between a point on the Earth's surface and the geoid. It is calculated as the integral of gravity along the plumb line from the geoid to the point of interest.

Geopotential numbers are used in precise geodesy to relate heights to the Earth's gravity field in a physically meaningful way. They provide a measure of height that is independent of local gravity variations.

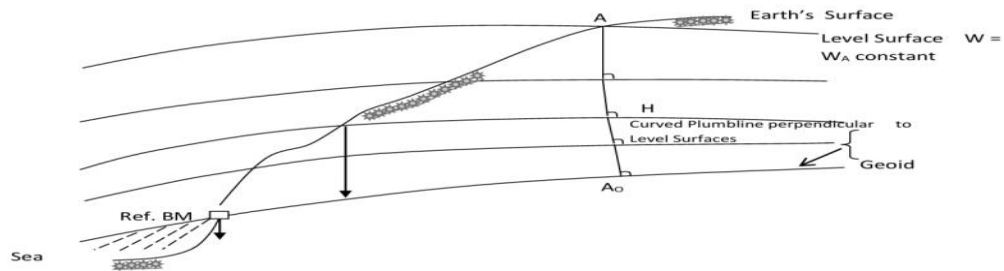
The geoid is usually considered the proper surface from which to reckon geodetic heights because it fulfils the criteria of water flow and nominally resides at mean sea level. Sea level, itself, does not exactly match the geoid because of the various physical factors. Therefore actually finding the geoid in order to realise a usable vertical datum is currently not possible from mean sea level measurements. Ideally, one would measure potential directly in some fashion analogous to measuring gravity acceleration directly. If this were possible, the resulting number would be a geopotential number. In other words, a geopotential number is the potential of the earth's gravity field at any point in space

Geopotential defines hydraulic head. Therefore if two points are at the same geopotential number, water will not flow between them due to the gravity alone. Geopotential decrease linearly with distance from the centre of the earth. It means the potential decreases with the distance away from the earth, so this difference is the positive number. This makes it a natural measure of distance. It does not depend on the path taken from the earth's centre to the point of interest. The magnitude of a geopotential number is less important than the relative values between two places. Therefore, we can define the geoid to have a geopotential number zero.

Geopotential numbers C_A are defined by the following equation $C_A = W_0 - W_A$

$$= - \int_{A0}^A dw = \int_{A0}^A g dh$$

which gives the change in gravity potential energy between a point on the geoid and another point of interest. The geopotential number for any place is the potential of the geoid W_0 minus the potential of that place W_A . C_A is called the geopotential number, which can be determined by observed geometrical height and observed Surface gravity.



Schematic Diagram for Level Surface

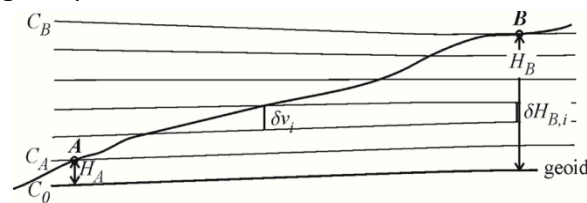
The geoid is the equipotential surface (W_0), that approximates mean sea level, g is the value of gravity at A and dh is the continuous function of height difference along the terrain from a point on the Geoid i.e. observed height difference. 'C' is measured in Geopotential units (gpu) where

$$1 \text{ gpu} = 1 \text{ kgal.m} = 1000 \text{ gal metre}$$

However geopotential numbers have units of energy, not length, and are therefore an 'unnatural' measure of height. Therefore Helmert Orthometric Heights most widely adopted nowadays are derived from Geopotential numbers as a natural measure of height for the Present Project.

1.10.3 Uncorrected Differential Levelling:

Levelling is a process by which the geometric height difference along the vertical is transferred from a reference station to a forward station. Suppose a levelling line connects two stations A and B as depicted in Figure 5, if the two stations are far enough apart, the levelling section will contain several turning points, the vertical geometric separation between which we denote as δv_i . Any two turning points are at two particular geopotential numbers, the difference of which is the potential gravity energy available to move water between them; hydraulic head. We also consider the vertical geometric separation of those two equipotential surface along the plumb line for B , $\delta H_{B,i}$.



Schematic Diagram for Geometrical Height and Orthometric Height

We will now argue that differential levelling does not, in general, produce orthometric heights. Figure 5 depicts two stations A and B , indicated by open circles, with geopotential numbers C_A and C_B , and at orthometric heights H_A and H_B , respectively. The geopotential surfaces, shown in cross section as lines, are not parallel; they converge towards the right. Therefore, it follows that $\delta v_i \neq$

$\delta H_{B,i}$. The height difference from A to B as determined by differential levelling is the sum of the δv_i . Therefore, because $\delta v_i \neq \delta H_{B,i}$ and the orthometric height at B can be written as $H_B = \sum \delta H_{B,i}$, it follows that $\sum \delta v_i \neq H_B$. Then we conclude that difference between differential levelling and orthometric heights is due to the role of gravity in height.

The differential levelling height differences differ from orthometric height differences by the amount that the surface gravity differs from gravity along the plumb line at that geo-potential. An immediate consequence of this is that two different levelling lines starting and ending at the same station will, in general, provide different values for the height of final station. This is because the two lines will run through different topography and, consequently geo-potential surface with different separations. Uncorrected differential levelling heights are not single valued, meaning the result you get depends on the route you to get there.

1.10.4 Orthometric Height-The Practical Elevation above Mean Sea Level

Orthometric height is the vertical distance from a point on the Earth's surface down to the geoid, representing mean sea level. It represents the "height above sea level" commonly used in maps, engineering drawings, and everyday applications.

Follows Gravity Direction: Orthometric height accounts for gravity variations by measuring along the plumb line—the direction gravity pulls.

National Vertical Datums: It is the height system used in most national vertical datums, such as NAVD88 in the USA and CGVD2013 in Canada.

Real-World Example: Mount Everest's orthometric height is approximately 8,848 m above mean sea level.

Orthometric height is the distance between the geoid and a point measured along the plumb line and taken positive upward from the geoid. In one sense, orthometric heights are purely geometric: they are the length of particular curve (a plumb line). However the curve depends on gravity in two ways. First, the curve begins at the geoid. |Second, plumb lines remains everywhere perpendicular to equi-potential surfaces through which they pass, so the shape of the curve is determined by the orientation of the equi-potential surfaces. Therefore, orthometric heights are closely related to gravity in addition to being a geometric quantity. Orthometric heights are directly related to geopotential number. We know that geopotential numbers are the difference in potential between Geoid W_0 and a point of interest A, W_A :

$$C_A = W_0 - W_A \quad C_A = \int_0^{H_A} g dH$$

Here it is understood that g is not a constant. Then from equation (8) we can derive the desired relationship:

$$C_A = \bar{g} H_A$$

$$H_A = \frac{C_A}{\bar{g}}$$

Meaning that a geopotential number is equal to an orthometric height multiplied by the average acceleration of gravity along the plumb line. Orthometric heights are likewise single valued, being a scaled value of a geopotential number.

If Orthometric heights are single valued it is logical to enquire whether surfaces of constant orthometric height form equipotential surfaces. The answer to this is unfortunately no. Consider the geopotential number of two different places with the same orthometric height. If orthometric heights formed equipotential surfaces, then two places at the same orthometric height must be at the same potential. Under this hypothesis equation (10) requires that the average gravity along the plumb lines of these different places necessarily be equal. However the acceleration of gravity depends on height, latitude, and the distribution of masses near enough to be of concern: it is constant in neither magnitude nor direction. There is no reason that the average gravity would be equal and in fact, it typically is not.

Therefore two points of equal orthometric height need not have the same gravity potential energy, meaning that they need not be on the same equipotential surface and, therefore, not at the same height from the perspective of geopotential numbers.

1.10.5 Normal Height

Normal heights are heights measured with respect to a normal gravity field, not the actual Earth gravity field.

Characteristics

- Based on a reference ellipsoidal normal gravity
- Geopotential Number is calculated through Surface gravity measurement.
- Widely used in European height systems

$$H_N = \frac{C}{\bar{\gamma}}$$

Where: (H_N) = Normal height

(C) = Geopotential number

(γ) = Mean normal gravity along the plumb line

1.10.6 Dynamic Height:

The geopotential numbers when divided by a specific gravity value, which will change its unit from Kgal.metre to metre and the result so obtained, will be the **Dynamic Height**.

$$H^{\text{dyn}} = C_p / \gamma_0$$

Where γ_0 is the value of normal gravity at latitude 24° which is taken in first level network for India. Normal gravity is determined from a mathematical model approximating actual gravity. The dynamic height has no geometric meaning as it represent a scaled difference of potential. Two different places on the same

equipotential surface have the same dynamic height but generally do not have the same orthometric height. Thus dynamic heights are not “distance from the Geoid”.

1.10.7 Helmert Orthometric Height

The Helmert Orthometric Height is one of the most commonly used in practice and is defined as

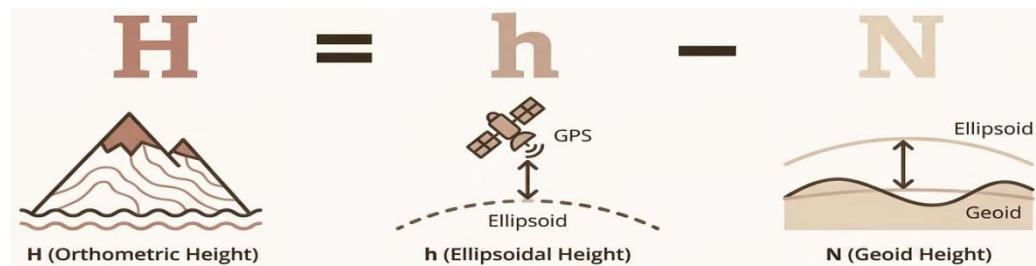
$$H_A = \frac{C_A}{g_A + 0.0424 H_A}$$

Where H_A is the computed orthometric Height in kilometres at the point 'A'.

C_A is the geo-potential number at the point A

g_A is the observed gravity in gals at the same point and

H_A is the height of the area of survey in kilometers.



1.11 Planning for Levelling Activity i.e. Manpower, Instrument and Duration

Planning is the most critical stage of any levelling activity.

Errors in levelling usually originate before observations begin, due to:

- Improper manpower deployment
- Inappropriate instrument selection
- Unrealistic estimation of daily progress

Well-planned levelling ensures:

- Accuracy
- Economy
- Safety

- Compliance with technical instructions

1.11.1 PLANNING OF MANPOWER

Standard Levelling Team Composition

Minimum manpower: 4 persons

High Precision Levelling (HPL): 5–6 persons

Leveller / Observer: Instrument setup, sighting, quality control

Staffman (BS): Holds staff on back sight point

Staffman (FS): Holds staff on fore sight point

Recorder / Computer: Field book entry, checks, summaries

Helper / Porter: Instrument carrying, BM marking

Supervisor (for HP work): Ensures compliance with tolerances

1.11.2 Skill Requirement of Manpower

Leveller: Knowledge of collimation, staff error, closure

Staffman: Vertical holding, shoe usage, barcode alignment

Recorder: Arithmetic accuracy, misclosure detection

Supervisor: Standards, tolerances, rejection criteria

1.11.3 PLANNING OF INSTRUMENTATION

Importance of Instrument Selection

The type of levelling instrument selected directly governs:

- Achievable accuracy
- Speed of observation
- Reliability of results
- Compliance with survey standards

Fundamental rule: *The accuracy of levelling cannot exceed the capability of the instrument–staff system used.* Therefore, instrument selection must be based on the purpose of levelling, not convenience.



Classification of
Levelling Instruments
Based on Work Type

Type of Levelling	Instrument Used	Application
Ordinary / Construction Levelling	Auto Level	Building works, roads, drainage
Control Levelling	Digital Level	Engineering control, mapping
High Precision Levelling (HPL)	Digital Level + Invar Staff	Datum definition, geodetic control

Auto Level (for Ordinary / Construction Levelling)

Characteristics

- Optical reading through telescope
- Manual staff reading by observer
- Automatic compensator maintains horizontal line of sight

Advantages

- Simple to operate
- Robust in rough field conditions
- Economical

Limitations

- Reading depends on observer skill
- Booking errors possible
- Personal bias unavoidable
- Not suitable for high-accuracy control

Digital Level (for Control & Precision Levelling)

A digital level automatically reads a barcode staff and stores data electronically.

Key Components

- CCD sensor
- Image processing unit
- Digital memory
- Barcode staff

Digital levels minimize:

- Reading errors – staff reading is automatic
- Booking errors – data stored electronically
- Personal bias – no subjective judgement.

For High Precision Levelling (HPL), the instrument alone is not sufficient. The staff material becomes equally critical.

Property	Invar Staff	Ordinary Aluminium Staff
Thermal expansion	Extremely low	High
Stability	Very high	Moderate
Suitability for HPL	Mandatory	Not allowed

1.11.4 PLANNING OF LEVELLING ROUTE

Route Selection Criteria

Importance of Route Selection

The levelling route is the physical path along which height transfer is carried out. Even with the best instrument and trained manpower, poor route selection will:

- Increase systematic errors
- Reduce accuracy
- Cause re-observation
- Affect long-term datum stability

Levelling accuracy is strongly dependent on the stability and environmental conditions of the route.

Stable Ground

Requirement:Ground should be firm, non-settling, and undisturbed

Preferred surfaces:Rock outcrops; Old masonry structures; Compacted natural ground

Avoid: Loose soil; Recently filled earth; Agricultural fields; Marshy or waterlogged areas

Unstable ground causes:Staff sinking; Turning point settlement; False height differences;

Minimal Traffic Interference

Why traffic is a problem: Vibrations disturb instrument setup; Staff holding becomes unstable; Safety risk to staff and equipment

Best practice: Avoid highways and busy roads during peak hours

If unavoidable:Work early morning; Use traffic cones and safety jackets

Traffic-induced vibration can cause sub-millimetre errors, which are unacceptable in control and precision levelling.

Good Intervisibility

Definition:Intervisibility means clear, unobstructed line of sight between:

Instrument; Back sight staff; Fore sight staff

Avoid:Curves and crests; Dense vegetation; Buildings and parked vehicles

Why important:Ensures accurate staff reading; Allows balanced BS–FS distances; Reduces refraction and pointing errors

Avoid Steep Gradients

Problems with steep gradients: Unequal sight lengths; Difficulty in staff verticality; Increased setup error

Recommended :Gentle slopes; Gradual elevation changes

Steep routes force long or short sights, violating the balanced sight principle

1.11.5 PLANNING OF DURATION (TIME ESTIMATION)

Factors Affecting Duration

Accurate time estimation is essential for:Work scheduling; Cost control ;Quality assurance; Avoiding rushed observations

The duration of levelling work depends on the following factors:

Terrain

Terrain Type	Effect on Duration
Plain / gentle slope	Faster progress
Undulating terrain	Moderate delay
Hilly / rocky terrain	Slow progress

Reasons: More frequent setups; Difficulty in staff placement;
Reduced intervisibility

Rule: The more uneven the terrain, the lower the daily progress.

Type of Levelling : Different classes of levelling demand different levels of care.

Type	Time Requirement
Ordinary levelling	Least
Control levelling	Moderate
High precision levelling	Maximum

Why high precision levelling takes longer: Double observations; Fore & back leveling; Strict tolerances; Frequent checks

Weather Conditions : Weather directly affects both observation quality and working hours.

Adverse conditions: High temperature → refraction; Rain → poor visibility & unsafe ground; Strong wind → staff instability

Best working window : Early morning; Mild, stable atmospheric conditions

Manpower Efficiency :

Manpower Quality	Impact
Trained, coordinated team	Faster & accurate
Untrained / new staff	Slow & error-prone

Efficient manpower ensures: Quick instrument setup; Proper staff holding
Fewer re-observations

Instrument Type

Instrument	Effect on Duration
Auto level	Faster but less reliable
Digital level	Slightly slower, more reliable
Digital + Invar staff	Slowest but highest accuracy

1.12 Different types of Bench Marks used by Survey of India.

1.12.1 Types of Benchmarks

The Standard Bench Marks were designed for erection in population centers, where the overcrowding of population leads to the constant removal of marks. The standard benchmarks are constructed / located at an average spacing of 25 to 30 km barring few exceptions. The standard benchmarks were also constructed at junctions of line of the Level network as well as near the tidal observatories. There are many types of Standard Bench Marks designed and used in survey of India.

Type 'P' Standard Bench Mark

Type 'M' Standard Bench Mark

Type 'A' Interred Bench Mark

Type 'B' Interred Bench Mark

Type 'C' Rock Cut Bench Mark

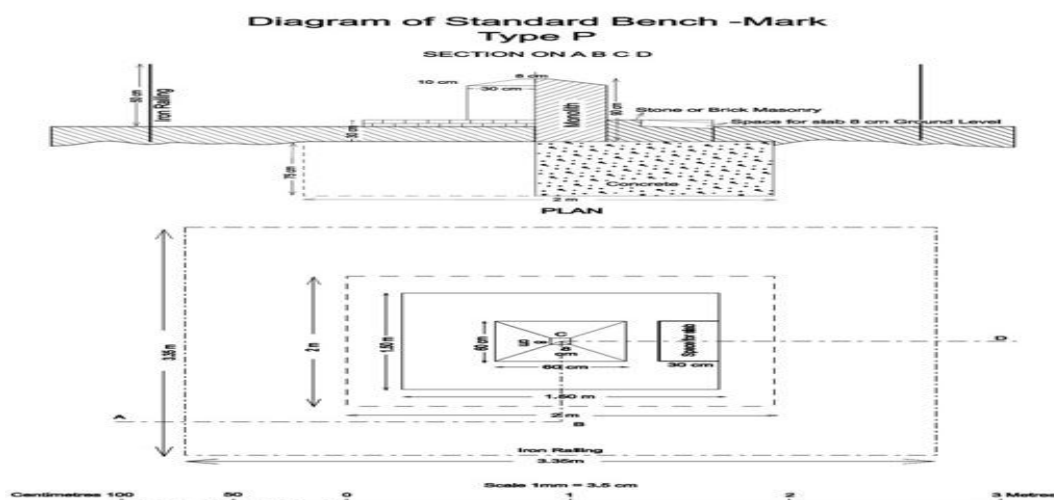
Type 'D' Interred Bench Mark

All above Bench Marks are stable and permanent in nature over with time.

1.12.2 Standard Bench Mark Type 'P':

Type 'P' bench mark is generally erected/constructed in big town or junction of HP levelling lines. It is important that only really good stone should be used for monolith and site for this type of bench mark have been selected with the help of regional Geological Survey India office.

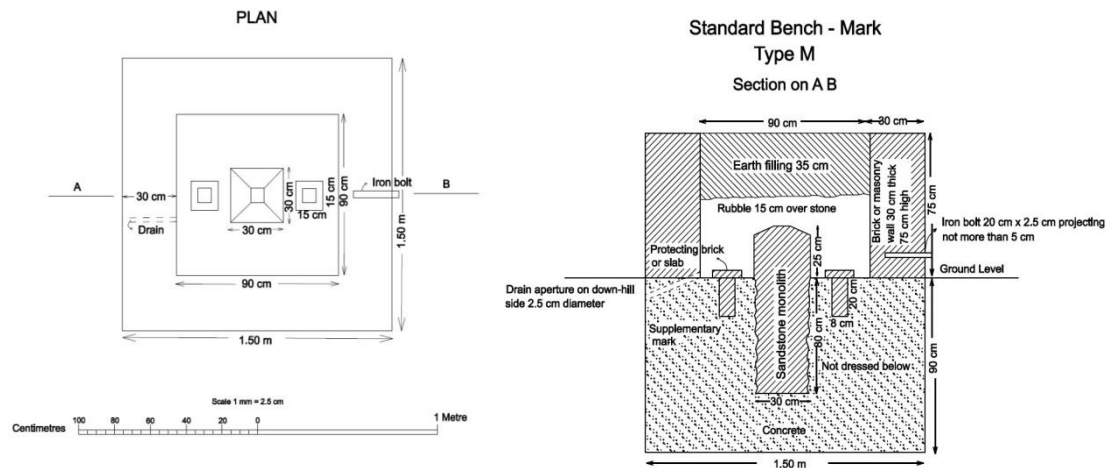
The standard benchmark type 'P' consists of a monolith 0.60m square at base and 0.90m high, the upper 0.10m being dressed to the form of a frustum of a pyramid terminating in a square of 0.08m side. The monolith rests on a bed of concrete 2m square, 0.75m deep and faced with stone or brick masonry 1.50m square and 0.30m deep, the upper surface of the monolith being about 0.60m above ground level. On one face of the monolith the inscription, is engraved.



1.12.3 Standard Bench-Mark (Type M) :

With a view to lessening expense, standards of different type are built in smaller towns and at the junction of HP levelling lines. The standard bench-mark (type M) consist of a monoliths of Chunār sandstone 0.30m square and 0.85m high of which 0.25m well dressed is above ground level and the remaining 0.60m in an undressed state is embedded in a bed of concrete

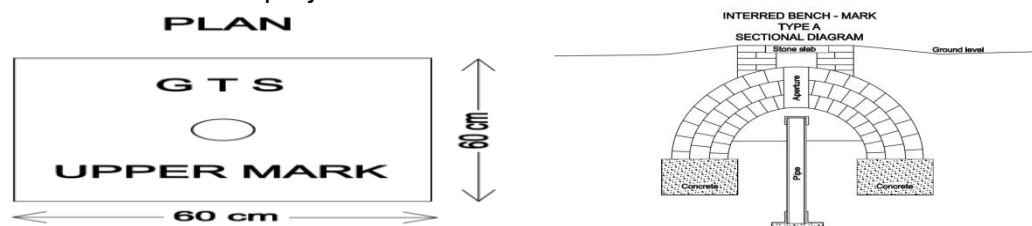
1.50m square and 0.90m deep. The upper 4 cm of the monolith are dressed to the form of a frustum of a pyramid, terminating in a square of 8cm side. The mark is enclosed in a masonry or brick wall 1.50m square outside, 0.30m thick and 0.75m high. There are three supplementary marks. Two of these consist of hard stone prisms 0.15m square, embedded flush with the level of the cement bed, and third is an iron bolt fixed horizontally in the wall of the enclosure, projecting about 5cm outwards.



1.12.4 Standard (Interred) Bench – Mark. Type A:

Type A bench-mark consists of a galvanized iron pipe 0.10 m in diameter and 0.90 m long standing on a slab of concrete 0.30-m square and 5 cm thick. The lower end of the pipe is provided with four flanges which rest on the slab of concrete. The upper surface of the slab is about 1.45 m below ground level, so that when fixed in position the top of the pipe is about 0.55m below the surface of the ground. The pipe is supported by well rammed earth up to 0.20 m from the top. A solid zinc cap is screwed and riveted to the upper end of the pipe and bears the inscription shown in the margin. A strongly built brick vault resting on concrete foundations covers and protects the bench – mark. Vertically above the pipe there is a 0.15-m square aperture in the crown of the vault and over this aperture there is a hollow masonry pillar carrying a mark-stone bearing the inscription shown in the margin.

The whole structure is below the surface of the ground except the upper mark – stone which projects a few cm above it.



1.12.5 Standard (Interred) Bench – Mark. Type B:

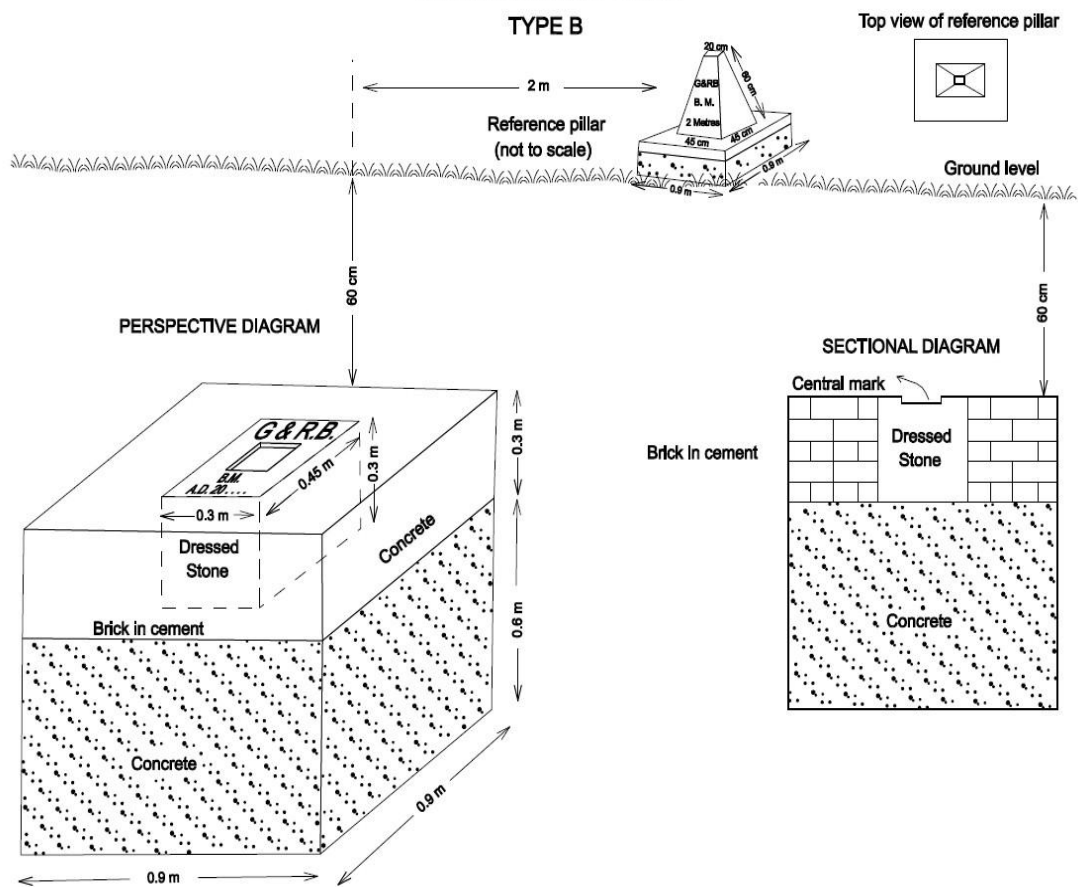
Due to non-availability of natural rocks in ground, this type of bench mark is constructed in Gangetic plane area with the help of dressed monolith of chunar stone flushed with cement concrete bed. The type B benchmark consists of a dressed stone $0.45\text{ m} \times 0.30\text{ m} \times 0.30\text{ m}$ set on a bed of concrete 0.90 m square and 0.60 m thick and surrounded by brick work set in cement.

The upper surface of the stone, which is $0.45\text{ m} \times 0.30\text{ m}$, is flush with that of the brick work and is about 0.60 m below ground level. The upper surface of the stone is inscribed as shown in the margin. The central mark is a 0.15 m square, and of 6 mm deep.

Where dressed stone $0.45\text{ m} \times 0.30\text{ m} \times 0.30\text{ m}$ is not available, a slab of cemented concrete of the same dimensions is substituted and a brass tablet is inserted in the centre flush with the level of the block. The brass tablet consists of a circular disk, 9 cm in diameter with a rod about 2 cm thick and about 9 cm long below it.

A reference pillar to indicate the place, where the bench mark lies buried, is essential. The pillar is built to a height of 0.90 m in the form of a frustum of a pyramid, 0.60 m square at base and tapering to about 0.15 m square at top, 2.0 m away from the place where the bench mark lies buried. The face of the reference pillar, opposite the bench mark should bear the inscription.

TYPE B



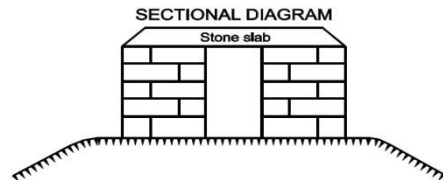
1.12.6 Standard (protectedRock-cut) Bench – Mark. Type C:

Rock Cut (Type C) bench marks are by far the most stable; thus a levelling detachment in the field may produce a primary protected bench mark more suitable than one built in anticipation of the levelling. Deeply engraved on solid ground rock with a circle of about 5 cm diameter and the letters are of about the same height. The inscription is protected by a hollow masonry pillar 0.60 square and 0.30 m high, on the top of which is fixed a stone bearing the inscription.

PLAN



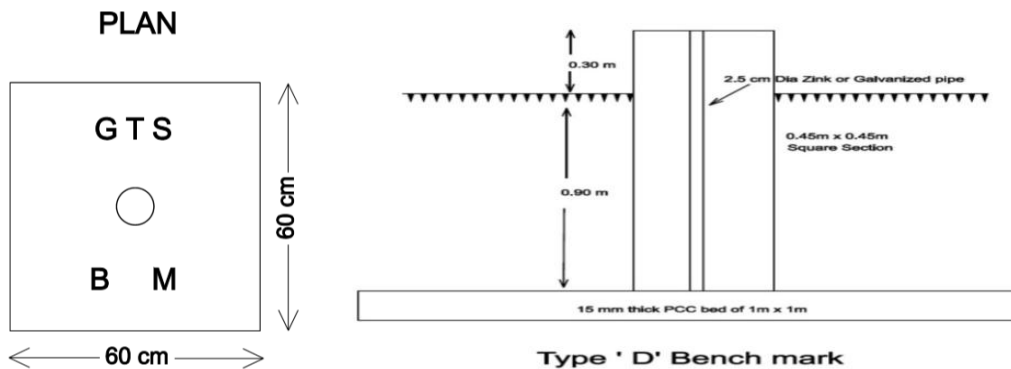
ROCK- CUT BENCH-MARK
TYPE C



1.12.7 Standard (Interred) Bench – Mark. Type D :

Consist of a pillar of cement concrete of dimension 0.45m square and 1.20m in height, of which 0.90m is below ground level, with a zinc pipe, 2.5 cm in

diameter and 1.20m in length embedded flush with the upper surface of the pillar. If zinc pipe is not available, hollow galvanized iron pipe may be used.



1.12.8 Inscribed Ordinary Bench Marks (OBMs) :

Usually consist of the inscription GTS-BOM or BOM or BM or OBM or BMO, Where O represent the point at which height of bench mark is measured. Between two SBM, ordinary bench marks will be inscribed which is not same design continuously. Most probable site for ordinary Bench Marks are

- Ground Rock
- Big Boulder
- Memorials And Tombs
- Church, Shrine or Temple
- Boundary Pillar
- Ancient Building
- Plinth or Floor of Government Office Like Post Office, Kacheri
- Government Circuit House / Guest House
- Tanks And Wells
- Railway Stations And Platforms
- Parapets of Girder Bridges
- Pedestals of Lamp Post
- Plinth of Culverts
- Plinth of Km Stone Pillar

OBMs are temporary in nature and constructed to provide necessary quality control during levelling work. These benchmarks are also used to close days work. However, these Benchmark may also be used for further extension of vertical control.

1.13 HIGH PRECISION LEVELLING (HPL)

High Precision Levelling (HPL) represents the highest standard of vertical control used in geodesy. It is designed to determine height differences with sub-millimetre accuracy over long distances. HPL is essential for establishing and maintaining national vertical datums, validating geoid models and supporting modern GNSS-based heighting systems.

Unlike ordinary engineering levelling, HPL requires strict observational discipline, specialised instrumentation and rigorous error control.

High Precision Levelling is required in situations where very small height differences must be determined reliably; Transfer of Mean Sea Level inland; Redefinition and maintenance of national height reference

1.13.1 Methodology of High Precision Levelling

Field Procedure

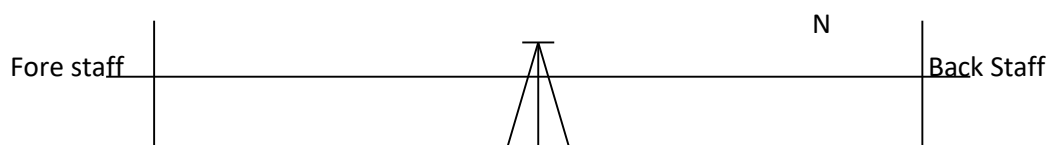
- The levelling line constitutes of sections of 1 km to 2 km apart, connecting two pucca Bench Marks by levelling.
- The instrument will be kept mid-way (within 1 metre) between fore and back staves, by careful pacing. This can also be checked by distance shown by Digital level.
- In digital level proper command is to be given to check the instrument position. Proper precautions must be taken to keep the instrument midway between the two staves.
- One particular foot screw will be kept on one particular leg of the stand and this leg will face one particular staff throughout the levelling operations.
- The length of sight depends upon the atmospheric conditions but should not exceed 60 m unless a wide river or any other obstacle is to be crossed.
- The staves reading are taken to four places of decimal of a metre.
- Two sets of observations with instrument height changed significantly (at least 2 cms.) for second set, will be taken at each instrument station. The two sets should agree within 0.0009 m. In case this limit is exceeded, continue observations till two consecutive sets agree within the prescribed limit.
- The order of observations for two sets will be Back-Fore and Fore-Back. Readings on the fore and back staves for a particular set should be taken in quick succession. The staves are erected upon heavy 'punja' (Iron shoe), which are provided with the instrument by the vendor.
- No staff reading should be taken on the bottom portion i.e. below 30 cm and above 270 cm from the foot of the staff, unless the staff is placed on a bench mark at least 30 cm above ground level.
- Even number of stations should be made between two consecutive bench marks to ensure that the same staff is placed on both the bench marks i.e. starting and closing bench mark of section.
- At the spot of work weather conditions will be recorded on data register. Starting time, Temperature at every hour and closing time with temperature will be recorded. It is better to take temperature at instrument height, and at the same height of staff.
- Bench marks must be established 1 to 2 km apart. As far as possible bench marks must be carefully chosen on permanent structure or rock-in-situ. Avoid parapets. Abutments / piers may be selected.
- The day's work should be closed on a bench mark or on a pucca mark.

- Observations should be avoided during sunset and sunrise to avoid the rapid change in atmospheric refractions at these times, which may affect the observations.
- To avoid sag in staves, these should be carried by the staff man keeping one particular side up and down alternatively. Sag should be checked fortnightly in the field.
- The back leveller will send the results section wise by email, Fax message and Whatsapp by scanning the format given below.

Sl. No.	BM No	Run Down Distance	Consecutive distance	Brief Description of BM	Run Down value Z	Consecutive dh

- Back levelling values will be compared against fore levelling values and if difference is more than permissible limits then relevelment instructions will be issued to the observer by email or telephone from HQ.
- No fore levelling value will be supplied to the Back Leveller from HQ to avoid the manipulation/ fudging.
- The discrepancy allowed between fore and back levelling is governed by the formulae $4\text{mm}\sqrt{K}$, where K is the distance between two bench marks in kilometers.
- If back value is not agreeing, relevelment will be done in fore direction. If after relevelment the values of fore are not in agreement, relevelment in back direction will be done. This process will be repeated until consecutive on fore value agrees with one back value within the prescribed limit.
- The direction of sun rays in relation to the position of staff to be shown correctly while sketching the diagram. Atmospheric temperature should be observed at starting and closing of observations and after every one hour during the observation.

✱



- The difference of fore and back values (F-B) should be checked continuously at PHQ, this difference should not be in one direction only. It may be sometime positive value and sometimes negative value so that $\Sigma (F-B)$ is minimized and is within permissible limit. If this does not happen, to control the F-B some sections to the judiciously selected for relevelment those are marginally in agreement.
- In a line, 20% re-levelment are accepted. Levelling line is to be revised in the portions where relevelments are more.
- There should be an interval of at least 2 months between fore and back levelling.

- If pucca structures are not available after one to three kilometers of levelling. A mark O or will be cut on suitable pucca structure.
- Description of Bench Mark will be written in Hindi & English giving full details of surroundings, i.e. distances & bearings from the centre of road, milestone and other permanent structure nearby it. It is important to give sheet number 1: 50,000 scale, in which Bench Mark falls. Coordinates of every bench mark should be recorded with the help of Mobile Phone/Hand held GPS.
- BM No. of starting Bench mark should be encircled with blue colour pencil and closing BM No, should be encircled with red colour pencil in the case of observation with N-3 level.
- Observed data with Digital level between two Bench Marks must be recorded in register. During recording the data, data must be verified twice. The form of data recorded in data register and Description Register is given below:

1.13.2 Check Levelling :

A line of H.P. levelling will start & close at a permanent bench mark previously established by H.P. Levelling line. The stability and identification of old bench marks or on which the work started or closed , must be verified by carrying out check levelling at the old bench marks. At least rundown of four (4) bench marks should agree with in the permissible limit i.e. $3\sqrt{k}$ mm where k is distance in kilometer.

The fact that B.M. used is identical with the one where description has been provided to the leveller must be categorically stated in the field book . Changes of description if any must be recorded.

Check levelling also be done when bench marks of another detachment are connected and also at the intermediate junction points where the new levelling line crosses an old one. The results of check levelling to be sent to HQ for approval of concerned Director, GDC.

1.13.3 FIELD SHEETS :

- Field Data Register and BM Description Register will be used for recording the data in the field.
- B.M. to B.M. difference of height, distance and coordinate (Observed by hand held GPS of Mobile Phones) of BMs will be Recorded in register.
- All the headings and titles of each page are filled before the observation are recorded. Fore or back direction will be written on the top of field sheet register along the name of the levelling line.
- All the bench marks will be serially numbered.
- Branch line bench marks will be numbered as 15/1, 15/2 so on if branch line starts from B.M. No. 15.

- In case of Branch Line a remark must be written in the data register against the bench marks from a branch line starts. "From this point emanates a branch line "and at the last point, where branch line ends "the Branch - line ends here."
- Relevelment of fore direction and of back direction will be clearly indicated in the Remarks column of the field register.
- At the starting and end of the section, description of the bench mark with sketch will be written. If old bench mark is connected, its old description must be verified and old Topo No. will be written at the bottom with remarks.
- "Identical with B.M. No.--- / (Sheet No.) old description holds good or revised" as the case may be.
- Geographical names must be verified from the latest available maps.
- The description of permanent bench mark, these located in congested city area, neat sketch showing the location of the bench mark must be drawn.
- For check levelling, separate field sheets will be used and the bench mark will be numbered serially along with their topo Nos. with remarks as given above.
- New observed difference of height must be sent to PHQ and approval of check levelling must be obtained from wing H.Q. before leaving the place for further work.
- The day's work will be checked by observer finally before carrying outfield computations.
- The field Register should be duly signed by the observer on every page. The first page and last page must contain full name of observer.
- Copying of original field data not to be done in any circumstances and it should be done at the working site only.
- The first page of field data register for check levelling must contain a certificate

1.13.4 Levelling Procedure

- Levelling is carried out in closed loops or double runs
- Forward and backward observations are compared
- Multiple setups are used to average out random errors
- Observations are recorded systematically and verified

Observation Conditions: Conducted during early morning or late evening; Avoids periods of strong sunlight and atmospheric turbulence; Stable weather conditions are preferred

Redundancy: Repeated observations; Multiple closures; Independent checks; Redundancy ensures detection and correction of errors.

1.13.5 Instrumentation Used in HPL

Digital Levels: High-resolution digital levels (0.1 mm or better); Automatic reading and data storage; Built-in error checks and calibration routines

Invar Staffs: Made of nickel-iron alloy; Very low coefficient of thermal expansion; Essential for sub-millimetre accuracy

Tripods and Accessories : Heavy, vibration-resistant tripods; Stable staff supports; Temperature sensors (in advanced surveys); Instrumentation quality is critical in achieving HPL accuracy.

Error Minimization in High Precision Levelling: Error minimization is the core principle of HPL.

Instrumental Error Control: Regular collimation tests; Two-peg test before and after survey; Calibration of staffs

Balanced sight lengths: Cancel collimation errors; Minimise curvature and refraction effects; Reduce systematic error accumulation